Hand Gesture Detection System — Command Reference

Total registered commands: **20**
Every command has a fixed numeric code starting from zero (suitable for simple output like serial/Arduino) as well as a human-readable text identifier.

Code	Command ID	Label	Label (alt)	Category	Description	Parameters
0	`finger.swipe.down`	Finger Swipe Down	Finger Swipe Down	**gesture**	While pointing, the index fingertip moves downward	`speed`: float
1	`finger.swipe.up`	Finger Swipe Up	Finger Swipe Up	**gesture**	While pointing, the index fingertip moves upward	`speed`: float
2	`finger.swipe.left`	Finger Swipe Left	Finger Swipe Left	**gesture**	While pointing, the index fingertip moves left	`speed`: float
3	`finger.swipe.right`	Finger Swipe Right	Finger Swipe Right	**gesture**	While pointing, the index fingertip moves right	`speed`: float
4	`hand.fist`	Fist	Fist	**gesture**	All four fingers (index through pinky) fully curl in	-
5	`hand.open`	Open Hand	Open Hand	**gesture**	The hand relaxes out of a fist back into an open pose	-
6	`hand.move.left`	Hand Move Left	Hand Move Left	**movement**	The palm center moves to the left	`speed`: float
7	`hand.move.right`	Hand Move Right	Hand Move Right	**movement**	The palm center moves to the right	`speed`: float
8	`hand.move.up`	Hand Move Up	Hand Move Up	**movement**	The palm center moves upward	`speed`: float
9	`hand.move.down`	Hand Move Down	Hand Move Down	**movement**	The palm center moves downward	`speed`: float
10	`hand.move.forward`	Hand Move Forward	Hand Move Forward	**movement**	The hand moves closer to the camera (based on the hand appearing larger)	`speed`: float
11	`hand.move.backward`	Hand Move Backward	Hand Move Backward	**movement**	The hand moves away from the camera (based on the hand appearing smaller)	`speed`: float
12	`hand.pinch.start`	Pinch Start	Pinch Start	**gesture**	The thumb tip and index tip come close together	-
13	`hand.pinch.rub`	Pinch Rub	Pinch Rub	**gesture**	While the two fingers are close together, they move back and forth (rubbing)	`intensity`: float
14	`hand.pinch.end`	Pinch End	Pinch End	**gesture**		-
15	`hand.thumb.show`	Thumb Shown	Thumb Shown	**gesture**	Only the thumb is extended, with the other four fingers curled in	`direction`: str
16	`hand.thumb.hide`	Thumb Hidden	Thumb Hidden	**gesture**		-
17	`hand.open_palm.swipe_right`	Open Palm Swipe Right	Open Palm Swipe Right	**gesture**	All five fingers are extended and the whole hand moves to the right	`speed`: float
18	`hand.rig.frame`	Full Hand Rig Frame	Full Hand Rig Frame	**rig**	The position of every one of the 21 hand landmarks (exactly matching MediaPipe's raw output) along with each finger's curl angle and extended/curled state, emitted on every frame (or at a capped rate). For connecting to a 3D model/character rig, robotics, or any consumer that needs the raw data.	`landmarks`: Array of 21 normalized [x, y, z] points, in standard MediaPipe order, `landmarks_named`: The same 21 points keyed by joint name (e.g. INDEX_TIP), `finger_states`: Boolean dictionary of extended/curled for each finger, `finger_curl_angles`: Each finger's curl angle in degrees (180 = fully straight), `palm_center`: [x, y, z] center of the palm, `hand_size`: Approximate hand size - usable to estimate depth/distance from the

						camera
19	`hands.pair.frame`	Two-Hand Frame Shape	Two-Hand Frame Shape	**pair**	When both hands are seen at the same time, a quadrilateral is built between reference points on the two hands (wrist and index finger base) that changes shape/size/angle as the hands move - exactly the ‘two-hand framing’ pattern seen in MediaPipe demos.	`corners` : Four corner points in order [bottom-left, top-left, top-right, bottom-right], each [x, y], `width` : Normalized distance between the two wrists (e.g. usable for zoom control), `height` : Average distance from wrist to index finger base for each hand, `center` : [x, y] center of the quadrilateral, `angle_deg` : Tilt angle of the line connecting the two hands, in degrees, `area` : Approximate area of the quadrilateral (normalized, between 0 and 1)